



Fire Prevention & Fire Fighting (FPFF)

59 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. In the event of a fire, the doors to stair tower must be closed to prevent the spread of fire by what?

- A) Convection
- B) Conduction
- C) Radiation
- D) Ventilation

Correct Answer: A (Convection)

Explanation: Closing fire doors prevents the spread of fire and smoke primarily through convection, which is the transfer of heat by the movement of hot gases and air. This compartmentalization is a key passive fire protection strategy as per SOLAS requirements.

Q2. The galley on your cargo vessel has an area of 2900 square feet, what will fulfil the requirements for fire protection?

- A) One B-I extinguisher
- B) One B-II extinguisher
- C) One B-II and one C-II extinguisher
- D) One B-V extinguisher

Correct Answer: C (One B-II and one C-II extinguisher)

Explanation: SOLAS regulations require specific fire protection for galleys based on area. For a galley of 2900 sq ft, the requirement is typically one Class B-II (medium size) and one Class B-V (large size) fixed foam system, or equivalent approved system. The options provided suggest a misinterpretation of extinguisher classes; typically, extinguishers are classified by type (A, B, C, etc.) and size, not specifically 'B-V' as a standalone extinguisher type.

Q3. The most likely location for a liquid cargo fire to occur on a tanker would be?

- A) In the midships house
- B) At the main deck manifold
- C) At the vent header
- D) In the pump room

Correct Answer: D (In the pump room)

Explanation: Pump rooms on tankers contain flammable liquids and ignition sources (e.g., electrical equipment, pumps), making them high-risk areas for liquid cargo fires. Proper fire detection and suppression systems, such as foam or inert gas, are critical in these spaces.



- A) Fuel
- B) Oxygen
- C) Heat
- D) All of Above

Correct Answer: D (All of Above)

Explanation: The fire triangle represents the three essential components required for fire to exist: fuel (combustible material), oxygen (an oxidizer), and heat (ignition source). Removing any one of these elements will extinguish or prevent a fire.

Q5. What kind of foam used in engine room?

- A) 1:500
- B) 1:1000
- C) 1:200
- D) 1:100

Correct Answer: B (1:1000)

Explanation: The standard dilution ratio for protein-based or synthetic film-forming (SFFP) foams used in engine rooms and for hydrocarbon fires on vessels is typically 1:1000 (foam concentrate to water). This ratio is crucial for effective fire suppression.

Q6. Dry Chemical Powder Extinguisher System is?

- A) Smothering
- B) Flame Inhibitor
- C) Fire Retardant
- D) None of Above

Correct Answer: B (Flame Inhibitor)

Explanation: Dry chemical powder (DCP) extinguishers primarily function by interrupting the chemical chain reaction of the fire, acting as a flame inhibitor. While they also have some smothering effect, their primary extinguishing mechanism is disrupting the combustion process.

Q7. Fire Plan Location is?

- A) Bridge
- B) Port and STBD side entrance
- C) Midships
- D) All of the above

Correct Answer: D (All of the above)

Explanation: Fire plans, mandated by SOLAS, are strategically posted at various accessible locations throughout the ship, including the bridge, main entrances (port and starboard sides), and near accommodation spaces like the midships area. This ensures crews and emergency responders can quickly locate vital information during a fire.

Q8. Class A Fire Extinguisher is best used on?

- A) Liquids
- B) Solids
- C) Gas
- D) Metals

Correct Answer: B (Solids)

Explanation: Class A fire extinguishers are designed for fires involving ordinary combustibles, which are typically solid materials like wood, paper, textiles, and rubber. They work by cooling the fuel and wetting it, preventing reignition.



- A) 15 M
- B) 20 M
- C) 30 M
- D) 40 M

Correct Answer: C (30 M)

Explanation: The Fireman's life line, used for rescuing personnel from a burning vessel, is a robust safety line that is typically 30 meters in length, allowing for safe descent or ascent in various emergency scenarios.

Q10. Which type of Extinguisher is danger to recharge onboard?

- A) DCP
- B) CO2
- C) FOAM
- D) None of Above

Correct Answer: B (CO2)

Explanation: Recharging CO2 (Carbon Dioxide) extinguishers onboard a vessel is considered dangerous due to the high pressures involved and the risk of overfilling or contamination. Specialized shore-based facilities are typically used for their safe recharging and testing.

Q11. What is the colour coding of portable Water Extinguisher?

- A) White
- B) Black
- C) Yellow
- D) Signal Red

Correct Answer: D (Signal Red)

Explanation: Portable fire extinguishers are color-coded for quick identification. Water extinguishers, used for Class A fires (ordinary combustibles), are painted Signal Red as per SOLAS and other maritime safety regulations.

Q12. Full form of AFFF is "AQUEOUS FILM FORMING FOAM"

- A) True
- B) False

Correct Answer: A (True)

Explanation: AFFF stands for Aqueous Film Forming Foam, a highly effective firefighting foam concentrate used primarily for Class B fires (flammable liquids). It forms a film over the fuel surface, suppressing vapors and cooling.

Q13. Heat removal how it stops FIRE?

- A) Cooling
- B) Inhibition
- C) Starvation
- D) Smothering

Correct Answer: A (Cooling)

Explanation: Heat removal is a primary method of fire extinguishment. When applied to a fire, water absorbs heat and turns into steam, thereby lowering the temperature of the fuel below its ignition point. This process is known as cooling.



- A) 15 Min.
- B) 20 Min.
- C) 30 Min.
- D) 45 Min

Correct Answer: C (30 Min.)

Explanation: A standard 1200-liter capacity SCBA bottle is designed to provide approximately 30 minutes of breathable air at a normal breathing rate. This duration is crucial for safe escape and firefighting operations in oxygen-deficient environments.

Q16. What is the duration of EEBD?

- A) 5 Min.
- B) 8 Min.
- C) 10 Min.
- D) 12 Min.

Correct Answer: C (10 Min.)

Explanation: An EEBD (Emergency Escape Breathing Device) is designed for escape purposes only, providing a limited supply of air for approximately 10 minutes. This duration is sufficient to allow personnel to reach a safe area during an emergency.

Q17. Proper use of Fire Extinguisher "PASS"?

- A) Pull, Aim, Squeeze, Sweep
- B) Pass, Aim, Shut, Sit
- C) Pass, At, Sure, Speed
- D) Pass, Aim, Squeeze, Sweep

Correct Answer: A (Pull, Aim, Squeeze, Sweep)

Explanation: The PASS acronym is a mnemonic for the correct operation of most portable fire extinguishers: Pull the pin, Aim the nozzle at the base of the fire, Squeeze the operating lever, and Sweep the nozzle from side to side.

Q18. The visible range of Hand Flare in night time?

- A) 10 NM
- B) 25 NM
- C) 30 NM
- D) 40 NM

Correct Answer: A (10 NM)

Explanation: A handheld red distress flare, as specified by SOLAS, is designed to be visible for at least 10 nautical miles (NM) at night. Its bright red light is intended to attract attention from vessels or aircraft for rescue.

Q19. Practicable how often shall Rescue Boats to be launched with their assigned crew and Maneuvered in Water?

- A) Every Week
- B) Every Month
- C) Once in Three Month
- D) Every Year

Correct Answer: C (Once in Three Month)

Explanation: SOLAS regulations require that lifeboats and rescue boats be launched with their assigned crew and maneuvered in the water at least once every three months. This ensures the equipment is operational and the crew is proficient.



- A) Bridge
- B) Main Deck
- C) Engine Room
- D) None of Above

Correct Answer: D (None of Above)

Explanation: Automatic CO2 release systems are typically installed in high-risk machinery spaces such as engine rooms, boiler rooms, and paint lockers, where flammable liquids or large fires are a concern. These spaces are not typically the bridge or main deck.

Q21. In which language FIRE CONTROL PLAN is written Onboard?

- A) Russian
- B) Spanish
- C) English
- D) French

Correct Answer: C (English)

Explanation: Fire control plans, which detail the fire protection systems and equipment onboard, are required to be readily available and are usually written in English. This ensures clear communication and understanding by international crews and rescue services.

Q22. Class-C Fire Extinguisher is best used on?

- A) Solid
- B) Metal
- C) Electrical
- D) Gas

Correct Answer: C (Electrical)

Explanation: Class C fires involve energized electrical equipment. Extinguishing agents used for Class C fires must be non-conductive to prevent electrical shock hazards, making CO2 or dry chemical powder suitable choices.

Q23. What is the maximum weight of a Portable Extinguisher?

- A) 5 KG
- B) 9 KG
- C) 15 KG
- D) 23 KG

Correct Answer: D (23 KG)

Explanation: While regulations can vary, a common upper limit for the weight of a portable fire extinguisher that a single person can reasonably handle is around 23 kg. This ensures portability and ease of use during firefighting operations.

Q24. Foam Fixed Fire Fighting System is used in which type of Vessels?

- A) Oil Tanker
- B) Container Ship
- C) Gas Carriers
- D) Bulk Carriers

Correct Answer: A (Oil Tanker)

Explanation: Foam is a highly effective extinguishing agent for flammable liquid fires, making it a standard fixed firefighting system in cargo and slop tanks of oil tankers. It suppresses vapors and cools the surface.



- A) 350
- B) 400
- C) 225
- D) 300

Correct Answer: C (225)

Explanation: B-Class divisions, which relate to bulkheads and decks, are designed to prevent the passage of flame for a specified period. The maximum temperature allowed on the non-fire side is typically 75 degrees Celsius above ambient, meaning around 225 degrees Celsius.

Q26. Foam Extinguisher work on?

- A) Smothering
- B) Starvation
- C) Cooling
- D) Inhibition

Correct Answer: A (Smothering)

Explanation: Foam extinguishers work by forming a foam blanket over the surface of flammable liquids, which separates the fuel from the air, thus extinguishing the fire. This process is known as smothering.

Q27. How many hours diesel driven Fire Pump should run?

- A) 14 Hrs
- B) 16 Hrs
- C) 18 Hrs
- D) 20 Hrs

Correct Answer: C (18 Hrs)

Explanation: According to SOLAS, diesel-driven fire pumps must be capable of operating continuously for a minimum of 18 hours. This ensures a sustained water supply for firefighting operations during extended emergencies.

Q28. Which is the best Fire Extinguisher for Liquid Fire?

- A) CO2
- B) Water
- C) Foam
- D) DCP

Correct Answer: C (Foam)

Explanation: Foam extinguishers are considered the most effective for liquid fires (Class B) because they form a film that suppresses vapors and prevents re-ignition. Water would spread the burning liquid, and CO2/DCP may be less effective on large liquid surfaces.

Q29. Minimum Capacity of Emergency Fire Pumps?

- A) 10 m3/ Hrs
- B) 15 m3/ Hrs
- C) 20 m3/ Hrs
- D) 25 m3/ Hrs

Correct Answer: D (25 m3/ Hrs)

Explanation: SOLAS mandates that the minimum capacity of emergency fire pumps on vessels is 25 m³/hour. This ensures a sufficient flow rate of water for essential firefighting operations in critical situations.



- A) To provide a visual barrier against smoke
- B) To contain the spread of fire within compartments
- C) To enhance communication among firefighting teams
- D) To store firefighting equipment safely

Correct Answer: B (To contain the spread of fire within compartments)

Explanation: Fire-resistant bulkheads are a critical component of a ship's passive fire protection system. Their primary purpose is to prevent the spread of fire and hot gases between compartments, thereby containing the fire and providing time for evacuation.

Practice Set 2

Q31. A class - D Fire would involve the burning of:

- A) Diesel Oil
- B) Dunnage
- C) Magnesium
- D) Electrical Insulation

Correct Answer: C (Magnesium)

Explanation: Class D fires involve combustible metals like magnesium, titanium, or aluminum. These fires require specialized extinguishing agents designed to react controllably with the burning metal, unlike agents used for other fire classes.

Q32. Cooling is the method of fire fighting by limitation of:

- A) Molecular Chain reaction
- B) Temperature
- C) Fuel
- D) Air

Correct Answer: B (Temperature)

Explanation: Cooling is a fire extinguishing method that involves reducing the temperature of the burning material below its ignition point. Water is a common cooling agent, as it absorbs a significant amount of heat when it evaporates into steam.

Q33. Which of the following fuels has the lowest flash point?

- A) Candle Wax
- B) Kerosene
- C) Coal
- D) Diesel Fuel

Correct Answer: B (Kerosene)

Explanation: Flash point is the lowest temperature at which a volatile liquid can vaporize to form an ignitable mixture in air. Kerosene has a lower flash point than diesel fuel, coal (which is a solid), and candle wax, making it more easily ignitable.

Q34. Which ship shall be provided with International Shore Connection for supply of water as Extinguishing agent?

- A) Tanker
- B) All ship of 500 GT and above
- C) Passenger ship carrying more than 500 person
- D) Passenger ship of 1000 person and above



Explanation: SOLAS regulations require that all ships of 500 GT and above be provided with an International Shore Connection.

~~This facilitates the supply of water from shore-based fire fighting appliances to the ship's fixed fire main.~~

Q35. The basic fire theory says that a fire needs 4 main elements in order to start: Heat, Fuel, Oxygen and ____?

- A) Hydrogen Sulphide
- B) Carbon-dioxide
- C) A chain reaction
- D) Benzene

Correct Answer: C (A chain reaction)

Explanation: The fire tetrahedron, representing the four essential elements for fire to exist and sustain itself, includes Heat, Fuel, Oxygen, and a Chain Reaction. Removing any one of these elements will extinguish the fire.

Q36. Putting fire out by removing fuel is called inhibition.

- A) True
- B) False

Correct Answer: B (False)

Explanation: Removing fuel from a fire is known as starvation, not inhibition. Inhibition refers to interfering with the chemical chain reaction of the combustion process, typically achieved with agents like Halon (though phased out) or certain dry chemicals.

Q37. The letters A, B and C on the portable fire extinguisher represent what?

- A) How much chemical it holds
- B) The first 3 letters of the extinguisher
- C) Type of fire on which it can be used
- D) None of the above

Correct Answer: C (Type of fire on which it can be used)

Explanation: The letters A, B, and C on portable fire extinguishers indicate the class of fire on which they are suitable for use. Class A is for ordinary combustibles, Class B for flammable liquids, and Class C for energized electrical fires.

Q38. Fire Plan tells what action to take in emergency situation by?

- A) The shore fire man
- B) The crew
- C) The Paramedics
- D) None of the above

Correct Answer: A (The shore fire man)

Explanation: Fire control plans, often referred to as ship's fire plans, are prominently displayed on board and are primarily intended for the ship's crew. They provide essential information on fire detection, suppression systems, and emergency procedures for crew response.

Q39. Class K fire involves high fatty oils?

- A) True
- B) False

Correct Answer: A (True)

Explanation: Class K fires, also known as Class F in some regions, specifically involve combustible cooking oils and fats, such as high-fatty oils found in commercial kitchens. These require specialized extinguishing agents that create a saponification effect.



- A) Furnishing and Personal effects
- B) Cigarette smoking
- C) Defective electrical connection
- D) All of the above

Correct Answer: D (All of the above)

Explanation: Accommodation spaces present multiple fire hazards, including furnishings (Class A), personal effects, potential ignition sources like smoking materials and faulty electrical connections. Therefore, all these factors contribute to the overall fire risk.

Q41. Fire Patrol to be done on Passenger vessels carrying more than 36 passengers?

- A) True
- B) False

Correct Answer: A (True)

Explanation: According to SOLAS, passenger ships carrying more than 36 passengers must conduct fire patrols. These patrols are conducted by crew members to detect and combat fires in their early stages.

Q42. Electricity switch board fire with power on, what type of extinguisher to be used?

- A) DCP
- B) Water
- C) Foam
- D) Carbon-dioxide

Correct Answer: D (Carbon-dioxide)

Explanation: For fires involving energized electrical equipment, such as switchboards, a Carbon Dioxide (CO2) extinguisher is the preferred choice. CO2 is a non-conductive gas that displaces oxygen and cools the fire without leaving residue.

Q43. Substances involving wood, plastic and paper are classified as?

- A) Class A fire
- B) Class B fire
- C) Class C fire
- D) Class D fire

Correct Answer: A (Class A fire)

Explanation: Class A fires involve ordinary combustible materials such as wood, paper, textiles, and some plastics. These fires are typically extinguished by cooling with water or using agents that absorb heat and smother.

Q44. Fire is the process of combustion accompanied by the rapid release of heat and ____?

- A) Light
- B) Energy
- C) Destruction
- D) None of the above

Correct Answer: A (Light)

Explanation: Fire is a rapid oxidation process characterized by the release of heat and light. It is the visible manifestation of combustion, which is the rapid chemical combination of a substance with an oxidant, usually oxygen.

Q45. On fires involving burning liquids, which of the extinguisher is considered most effective?

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- A) Water
- B) Foam
- C) DCP
- D) CO2

Correct Answer: B (Foam)

Explanation: Foam is considered the most effective extinguishing agent for fires involving burning liquids (Class B). It forms a blanket that separates the fuel from the air, suppresses vapor release, and cools the surface, preventing re-ignition.

Q46. Portable extinguisher should not be more than 13 kg.

- A) True
- B) False

Correct Answer: B (False)

Explanation: False. While portability is key, SOLAS regulations permit portable extinguishers up to a certain weight, with common limits often cited around 23 kg, not 13 kg. This ensures they can still be effectively handled.

Q47. Fire tends to start in a switchboard due to short circuit, this is which class fire?

- A) Class A
- B) Class B
- C) Class C
- D) Class D

Correct Answer: C (Class C)

Explanation: Fires involving electrical equipment and wiring (with power on) are classified as Class C fires. Using non-conductive extinguishing agents is crucial to prevent electrocution hazards.

Q48. Firefighter should come back from the area,

- A) After 10 minute of firefighting
- B) Upon instruction from other firefighter
- C) Upon hearing the warning sound from SCBA
- D) None of the above

Correct Answer: C (Upon hearing the warning sound from SCBA)

Explanation: The audible alarm on an SCBA set typically indicates a low air supply warning, signaling the need to evacuate the hazardous area immediately. This is a critical safety feature for firefighters.

Q49. If restricted visibility encountered in an accommodation fire, you would

- A) Carry a torch
- B) Stand up straight to keep your head above smoke
- C) Use of a fan to blow smoke and flames
- D) Rig lifelines to assist a retreat

Correct Answer: D (Rig lifelines to assist a retreat)

Explanation: In heavy smoke conditions during an accommodation fire, it is essential to stay low to breathe cleaner air and to use lifelines or guides to maintain orientation and facilitate retreat. Standing upright would expose you to the hottest smoke.



- A) Weight
- B) Temperature
- C) Electrical non-conductivity
- D) Cost

Correct Answer: C (Electrical non-conductivity)

Explanation: When fighting Class C fires (electrical), the most important characteristic of the extinguishing agent is its electrical non-conductivity. This prevents the risk of electric shock to the firefighter operating the extinguisher.

Q51. Foam tends to extinguish fire by

- A) Chemically absorbing the oxygen
- B) Smothering the burning material
- C) Releasing large amount of CO2
- D) None of the above

Correct Answer: B (Smothering the burning material)

Explanation: Foam extinguishes fires primarily by smothering. It forms a vapor-suppressing blanket over the surface of the burning liquid, preventing oxygen from reaching the fuel and stopping the combustion process.

Q52. What is your first action on seeing fire?

- A) Go to the muster station
- B) Extinguish the fire immediately
- C) Try on alarm and try to make an initial attempt
- D) None of the above

Correct Answer: C (Try on alarm and try to make an initial attempt)

Explanation: Upon sighting a fire, the immediate and most crucial action is to raise the alarm to alert the crew. Following this, an initial attempt to extinguish the fire should be made if it is safe to do so and the fire is small.

Q53. What is the color coding of portable DCP type extinguisher?

- A) Black
- B) Yellow
- C) Blue
- D) Signal Red

Correct Answer: C (Blue)

Explanation: Portable Dry Chemical Powder (DCP) extinguishers are identified by their blue colored band, as per international standards for visual identification of fire extinguisher types. Other colors indicate different extinguishing agents suitable for specific fire classes.

Q54. Fire alarm is...?

- A) Continuous ringing
- B) Verbal order by master
- C) 7 short blast followed by 1 long blast
- D) None of the above

Correct Answer: A (Continuous ringing)

Explanation: The standard fire alarm signal on a vessel is a continuous ringing or sounding, signifying an emergency. Specific blast patterns usually indicate other signals like abandon ship or man overboard.



- A) CO2
- B) Foam
- C) Water
- D) DCP

Correct Answer: A (CO2)

Explanation: CO2 extinguishers are the best choice for electrical fires because the gas is non-conductive and leaves no residue, preventing damage to sensitive electrical equipment. Water and foam can conduct electricity, and DCP might damage electrical components.

Q56. Fire Extinguish system required for propelling machines and diesel-driven motor?

- A) Hyper Mist
- B) CO2
- C) DCP
- D) Foam

Correct Answer: A (Hyper Mist)

Explanation: Hyper Mist systems, a type of water mist system, are highly effective for machinery spaces containing propelling machinery and diesel motors. They use a fine mist of water to cool and suppress fires with minimal water discharge.

Q57. Class-B Bulkhead are classified as B15/B0

- A) True
- B) False

Correct Answer: A (True)

Explanation: The classification 'B15' for bulkheads indicates that they are constructed to resist fire for 15 minutes. 'B0' signifies that the bulkhead provides integrity but not necessarily a specific fire resistance period, serving as a fire barrier.

Q58. DCP Fixed Fire Fighting system is used in which type of vessel?

- A) Oil Tankers
- B) Chemical Tankers
- C) Bulk Carriers
- D) Gas Carriers

Correct Answer: D (Gas Carriers)

Explanation: Dry Chemical Powder (DCP) fixed fire fighting systems are primarily installed in cargo spaces of Gas Carriers. These systems are effective against flammable gas fires and are chosen for their rapid knockdown capability.

Q59. If passengers are onboard when an Abandon Ship drill is carried out, they should_____?

- A) Watch
- B) Stay out of the way and do what they want
- C) Take Part
- D) Go to their quarters

Correct Answer: C (Take Part)

Explanation: During an abandon ship drill, passengers should actively participate under the crew's guidance. This ensures they understand emergency procedures and can respond appropriately if a real emergency occurs.



- A) To be most effective, Foam should be directed at the base of the fire
- B) Foam conducts electricity
- C) Foam is most effective on burning liquids which are flowing
- D) Foam can only be used to extinguish Class A Fire

Correct Answer: B (Foam conducts electricity)

Explanation: Firefighting foam is electrically conductive. It is most effective on Class B fires (flammable liquids) by forming a blanket that smothers the flames and prevents re-ignition, not for Class A fires.